Detecting changes in mixed-sampling rate data sequences

Joint work with Aaron Lowther & Idris Eckley

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CIRM Sept 2022

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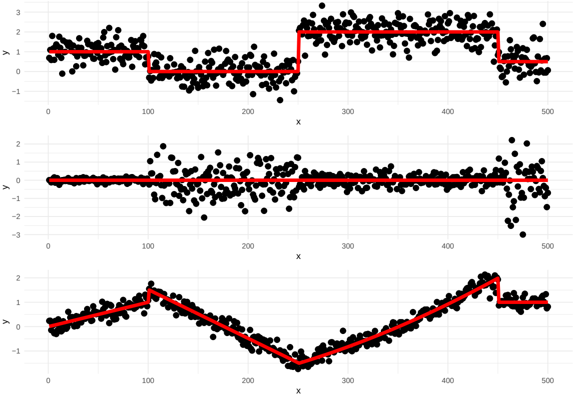
Outline

- Motivation
- Method
- Simulations
- Application

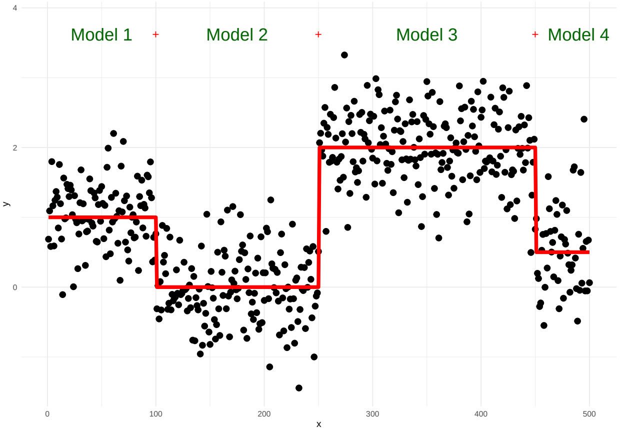
What are changepoints?

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

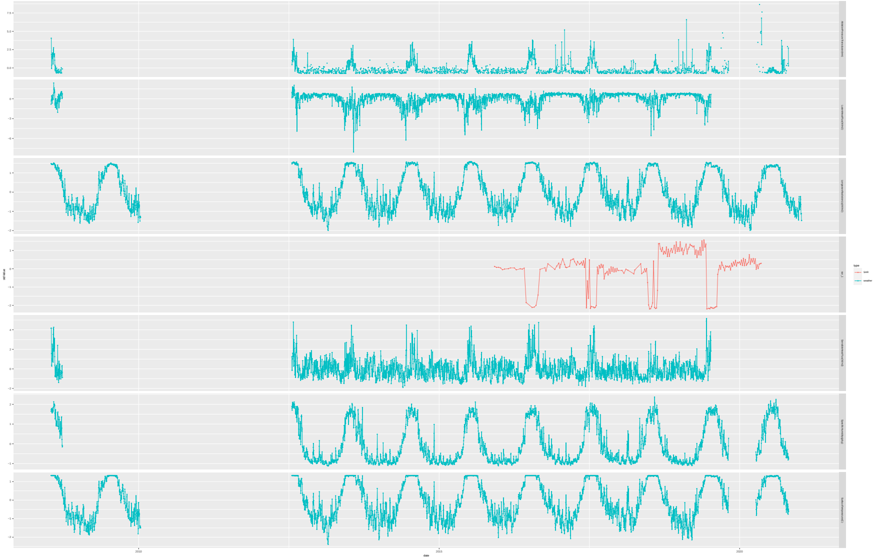
There are many different types of change.



Problem



Motivation Ice



Aim

Combine data collected using different sampling rates to detect changes in all series

- No subsampling / aggregation
- No interpolation

Problem: Existing techniques assume

- regular sampling rates
- observations across variables are observed at the same time

Method (Independence)

Key points:

- Likelihood doesn't assume regular sampling rate
- Create an observed time index across variables
- Use all data available up to a point to detect change
- Respect minimum segment lengths per series
- Respect penalties grow with sample size

Method (Independence)

Usually see:

$$\mathcal{C}\left(\mathbf{y}_{u:v}\right) = \sum_{p=1}^{P} \mathcal{C}_p\left(y_{p,u:v}\right),$$

where $u : v = \{u+1, u+2, \ldots, v\}$.

Now we have:

$$\mathcal{C}\left(\mathbf{y}_{(u,v]}\right) = \sum_{p=1}^{P} \mathcal{C}_p\left(y_{p,(u,v]}\right),$$

where $(u, v] = \{t \in (u, v]\}$.

Minimum segment length

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$\mathcal{C}(y_{p,(u,v]})$ may not be defined if $|y_{p,(u,v]}| < L_p$.

Naive solution: restrict to feasible locations.

Instead:

$$\tilde{\mathcal{C}}_p\left(y_{p,(u,v]}\right) = \begin{cases} \mathcal{C}_p\left(y_{p,(u,v]}\right) & \text{if } \left|y_{p,(u,v]}\right| \geq L_p \\ 0 & \text{otherwise} \end{cases} \qquad \text{for } p = 1, \ldots, P.$$

MSR cpts

$$\min_{\tilde{\tau},M} \left\{ \sum_{m=1}^{M+1} \left[\tilde{\mathcal{C}} \left(\mathbf{y}_{(\tau_{m-1},\tau_m]} \right) + \beta \right] \right\},$$

- $$\tilde{\tau} = \{[\tau_0, \tau_1, \dots, \tau_{M+1}] : \exists\, p \in \{1, \dots, P\} \text{ where } \left| y_{p,(\tau_m,\tau_{m+1}]} \right| \geq L_p$$
  $$\text{for } p = 1, \dots, P \text{ and } m = 0, \dots, M\}.$$
- $\tilde{\mathcal{C}}(\cdot) = \sum_{p=1}^{P} \tilde{\mathcal{C}}_p(\cdot)$
- $\beta = \sum_{p=1}^{P} \beta_p$

MSR-PELT

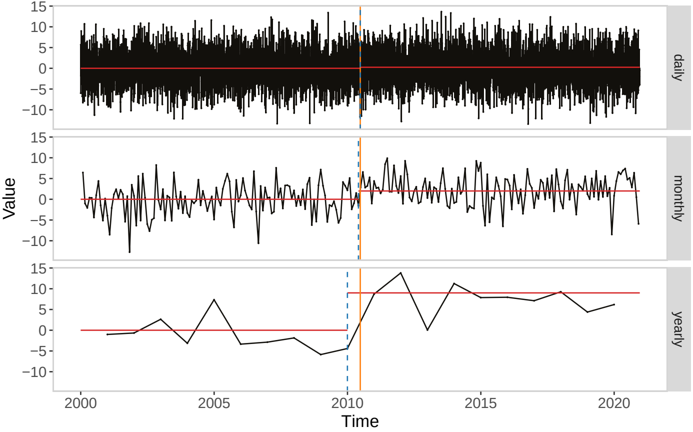
Solve using PELT. For $t > s$,

$$F(t) = \min_{s} \left\{ F(s) + \tilde{\mathcal{C}} \left(\mathbf{y}_{(s,t]} \right) + \beta \right\}.$$

Pruning the set s akin to Killick et al. (2012) remains valid for the new sets.

Power

Daily-Monthly-Yearly observations. $\sigma^2 = 4$, $\mu_1 = (0.25, 2, 9)$.



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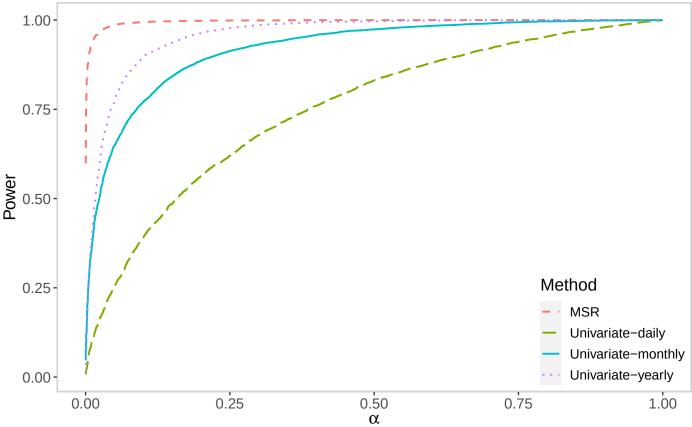
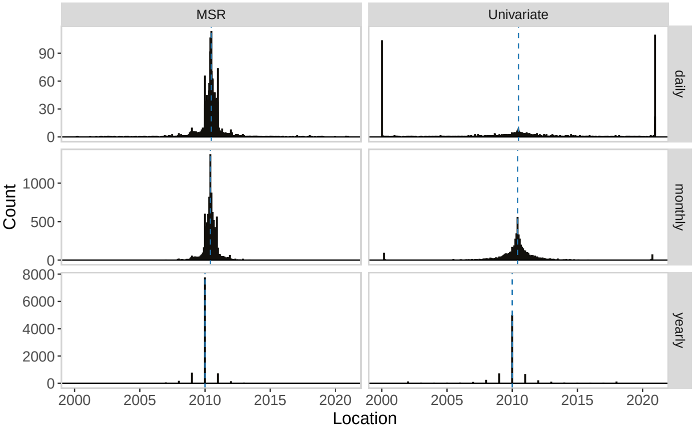


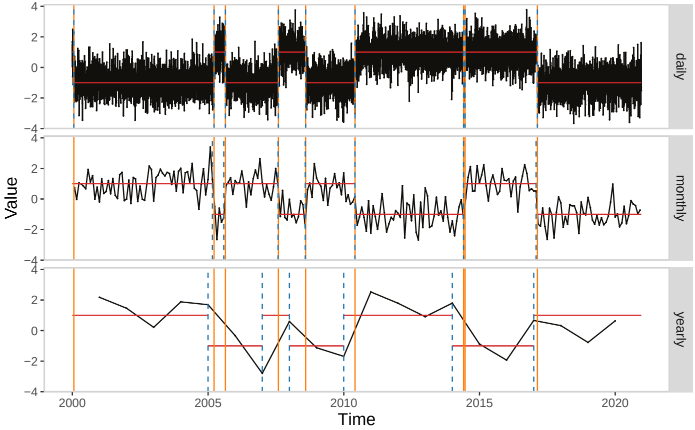
Location

Daily-Monthly-Yearly observations. $\sigma^2 = 4$, $\mu_1 = (0.25, 2, 9)$.



Multiple Changes

Daily-Monthly-Yearly observations. $\sigma^2 = 4$, Alternating mean $(1, -1)$.



Multiple Changes

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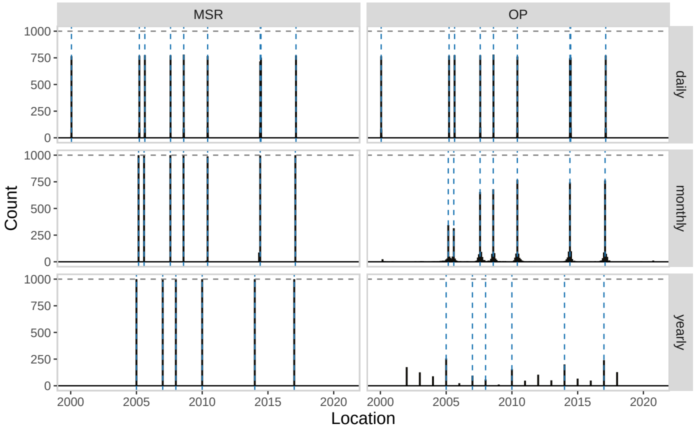
Daily-Monthly-Yearly observations. $\sigma^2 = 4$, Alternating mean $(1, -1)$.



Multiple Changes

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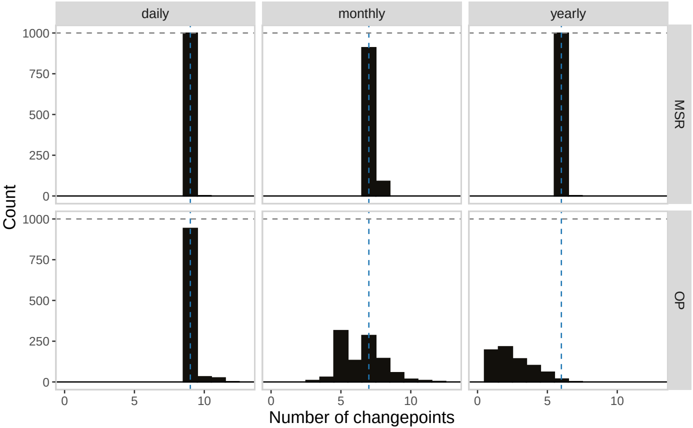
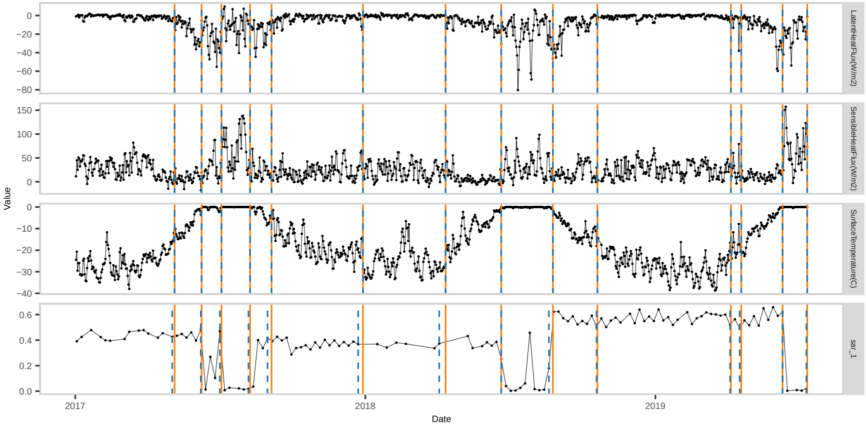


Application



Conclusion

- Able to detect changes in multivariate sequences with mixed sampling rates
- Also handles missing data within sequences
- Generic method that could be used with other test statistics

Next steps:

- Incorporate temporal dependence
- Incorporate cross-correlation

Detecting changes in covariance via random matrix theory

Joint work with Sean Ryan

Rebecca Killick(r.killick@lancs.ac.uk)
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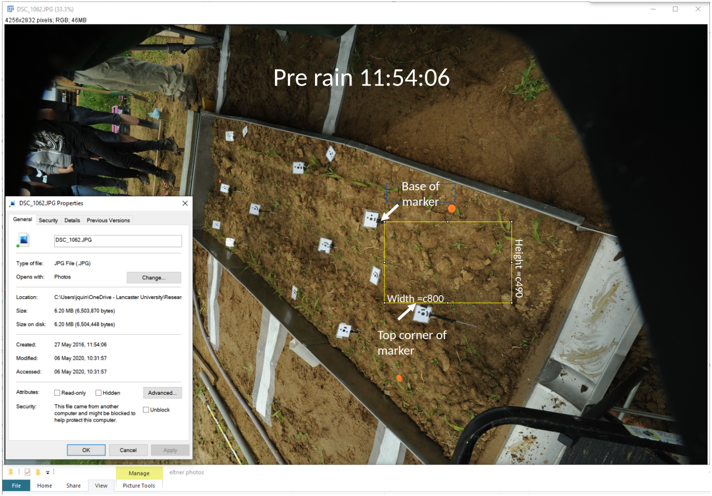
Outline

- Motivation
- Inspiration
- F-matrix solution
- Simulations
- Application

Motivation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Accessed: 06 May 2020, 10:31:57
Attributes: Read-only Hidden Advanced...
Security: This file came from another computer and might be blocked to help protect this computer.
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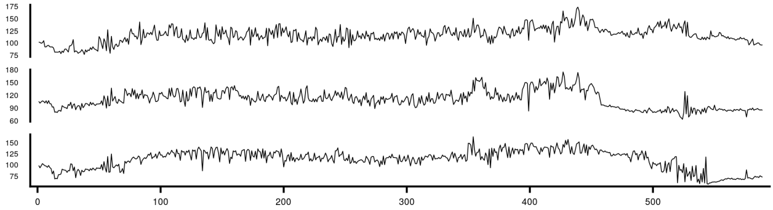
Motivation

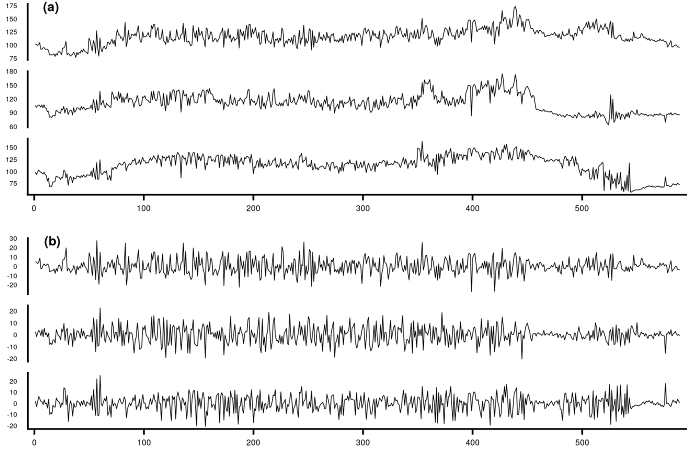
Motivation

Motivation


175
150
125
100
75
180
150
120
90
60
150
125
100
75
0
100
200
300
400
500

Motivation

Bayesian Trend Filter with Dynamic Shrinkage (Kowal et al. 2019), DSP R package



Aim

Detect multiple changes in covariance structure.

Existing Work

$$
\begin{aligned}
H_0 :& \Sigma_0^* = \Sigma_{1,p} = \cdots = \Sigma_{n,p} \\
H_1 :& \Sigma_1^* = \Sigma_{1,p} = \cdots = \Sigma_{\tau,p} \neq \Sigma_{\tau+1,p} = \cdots = \Sigma_{n,p} = \Sigma_2^*,
\end{aligned}
$$

where τ is unknown.

Typically test statistics,

$$\psi_{\tau,1}\bar{\Sigma}(0,\tau) - \psi_{\tau,2}\bar{\Sigma}(\tau,n),$$

where $\{\psi_{\tau,1}\}_{\tau=\ell+1}^{n-\ell}$ and $\{\psi_{\tau,2}\}_{\tau=\ell+1}^{n-\ell}$ are sequences of normalizing constants.

Large spectral norm $\implies$ changepoint.

Aue et al (2009), Galeano and Pena (2007), Wang et al (2017) ....

All need estimate of Σ_0 for thresholds!

Inspiration

Univariate: use functions of $\frac{\sigma_1}{\sigma_2}$.

Advantages:

- More natural view, 1 under the null, deviations suggest changes
- No need to estimate the variance under the null
- Scale invariant

Extension to multivariate

$$R(A, B) := B^{-1}A,$$

where $A, B \in \mathbb{R}^{p \times p}$.

$$T(A, B) = \sum_{j=1}^{p} \left(1 - \lambda_j(R(A, B))\right)^2 + \left(1 - \lambda_j^{-1}(R(A, B))\right)^2,$$

where $\lambda_j(R(A, B))$ is the jth largest eigenvalue of the matrix $R(A, B)$.

Extension to multivariate

$$T(\Sigma_1, \Sigma_2) = \sum_{j=1}^{p} \left(1 - \lambda_j(R(\Sigma_1, \Sigma_2))\right)^2 + \left(1 - \lambda_j^{-1}(R(\Sigma_1, \Sigma_2))\right)^2,$$

- T is symmetric i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_2, \Sigma_1)$;
- T is symmetric with respect to inversion of matrices i.e. $T(\Sigma_1, \Sigma_2) = T(\Sigma_1^{-1}, \Sigma_2^{-1})$;
- T does not depend on the underlying covariance.

If $\Sigma_1 = \Sigma_0 Z_1^T Z_1 \Sigma_0$ and $\Sigma_2 = \Sigma_0 Z_2^T Z_2 \Sigma_0$, then $T(\Sigma_1, \Sigma_2) = T(Z_1^T Z_1, Z_2^T Z_2)$.

Theory

The sample sizes n_1, n_2, and the dimension p grow to infinity such that

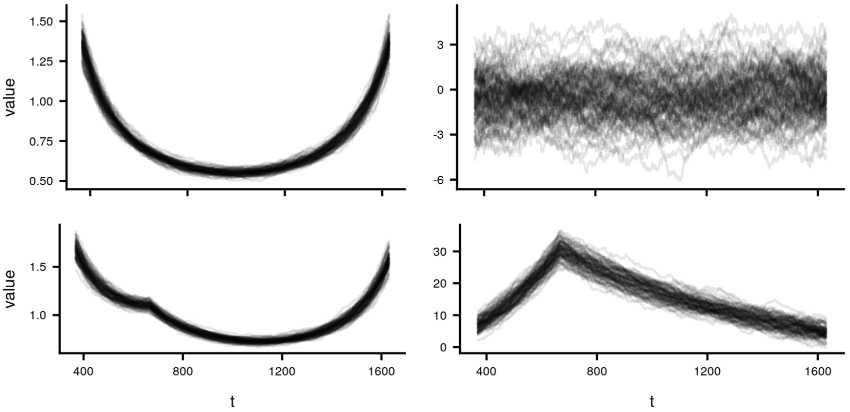
$$\gamma_{n_1} := \frac{p}{n_1} \to \gamma_1 \in (0,1),\ \gamma_{n_2} := \frac{p}{n_2} \to \gamma_2 \in (0,1) \text{ and } \boldsymbol{\gamma} := (\gamma_1, \gamma_2).$$

Then under some mild assumptions:

$$T\left(\frac{1}{n_1}X_{n_1,p}^T X_{n_1,p}, \frac{1}{n_2}X_{n_2,p}^T X_{n_2,p}\right) - p\int f^*(x)dF_{\boldsymbol{\gamma}}(x) \to N(\mu(\boldsymbol{\gamma}), \sigma^2(\boldsymbol{\gamma}))$$

Correction

$$\tilde{T}(t) := \sigma^{-1/2}(\gamma_{t/n})\left(T(\bar{\Sigma}(0,t),\bar{\Sigma}(t,n)) - p\int f^*(x)dF_{\gamma_{t/n}} - \mu(\gamma_{\tau/n})\right),$$



Single changepoint small

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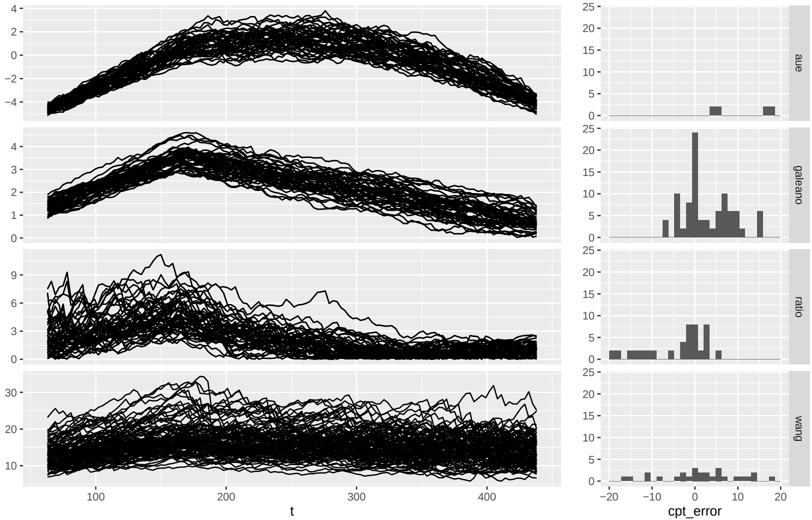
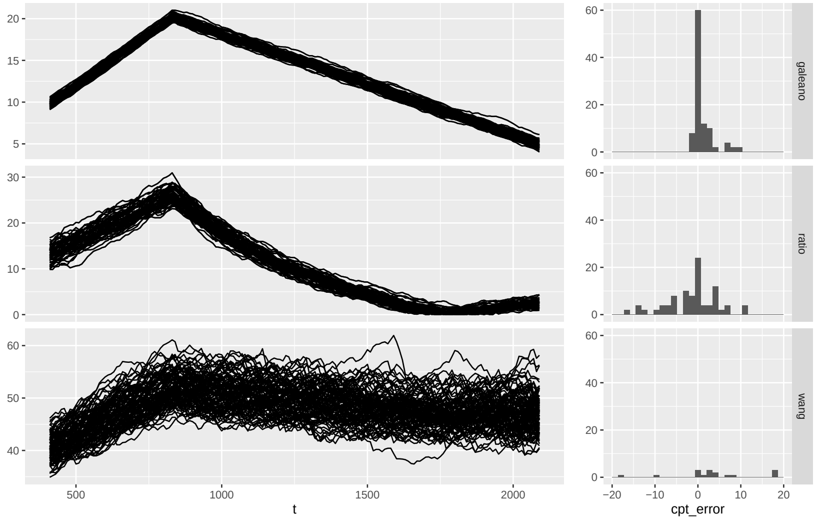


Single changepoint large



Multiple changepoints

p	n	Aue	Ratio	Wang	Galeano
		FPR			
3	500	0.36	0.24	0.63	**0.00**
	1000	0.48	0.28	0.77	**0.02**
	2000	0.54	0.31	0.85	**0.09**
	5000	0.59	0.31	0.90	**0.23**
10	500	0.25	0.16	0.27	**0.00**
	1000	0.32	0.13	0.43	**0.04**
	2000	0.34	0.10	0.53	**0.09**
	5000	0.37	**0.09**	0.62	0.21
30	2000		**0.02**	0.32	0.09
	5000		**0.02**	0.31	0.22
100	5000		**0.00**	0.45	0.26

Multiple changepoints

		TPR			
p	n	Aue	Ratio	Wang	Galeano
3	500	**0.55**	0.38	0.51	0.00
	1000	**0.59**	0.48	0.48	0.01
	2000	**0.63**	0.60	0.44	0.03
	5000	0.65	**0.68**	0.45	0.06
10	500	**0.56**	0.55	0.51	0.01
	1000	0.69	**0.75**	0.54	0.01
	2000	0.76	**0.87**	0.58	0.04
	5000	0.80	**0.91**	0.61	0.07
30	2000		**0.98**	0.45	0.05
	5000		**0.99**	0.56	0.10
100	5000		**1.00**	0.31	0.13

Multiple changepoints

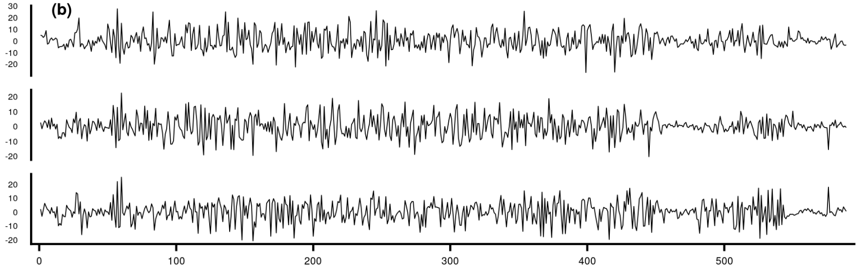
		MAE			
p	n	Aue	Ratio	Wang	Galeano
3	500	**26.05**	26.52	42.82	55.67
	1000	18.44	**15.36**	36.81	48.97
	2000	12.78	**7.65**	32.14	43.95
	5000	7.95	**2.93**	23.85	36.66
10	500	304.00	**303.63**	316.80	780.50
	1000	167.59	**127.71**	243.70	678.34
	2000	99.52	**46.44**	186.26	594.56
	5000	58.09	**16.62**	123.34	492.97
30	2000		**143.43**	1078.25	3079.91
	5000		**51.37**	525.14	2357.40
100	5000		**199.51**	7769.79	16281.43

Soil Application

Ratio: 64, 125, 184, 255, 340, 450, 527

Wang: 445 (huge eigenvalue affecting threshold)

Scientists: 64 dry - 76 very wet, increase 350, decrease 450



Summary

- New method based on ratio of covariance matrices
- Doesn't require estimation of Σ_0
- Good false postive control away from the boundaries
- Theoretical threshold works well for multiple changepoints

Future work

- Extend to mixed sampling rates and/or missingness